MultiConAD: A Unified Multilingual Conversational Dataset for Early Alzheimer's Detection

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Introduction

Background

- Dementia is a progressive cognitive syndrome with Alzheimer's disease (AD) as the leading cause. Conversation-based AD detection offers a cost-effective alternative to clinical methods, as language dysfunction is an early biomarker of AD.
- Most prior research has framed AD detection as a binary classification problem, limiting the ability to identify Mild Cognitive Impairment (MCI), a crucial stage for early intervention.
- Studies also primarily rely on single-language datasets, mainly in English, restricting cross-language generalizability.

Main contributions

1. Presenting a new multilingual conversational dataset for AD detection.
2. Performing fine-grained three-class classification (AD, MCI, HC).
3. Analyzing classification performance in monolingual and multilingual settings.

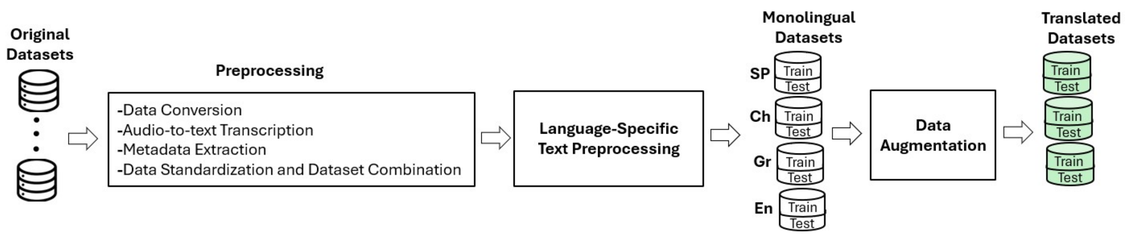
Dataset Creation

Sources

- Sixteen publicly available AD-related conversational datasets are unified to create a multilingual dataset in Chinese, Spanish, English, and Greek.
- An overview of all datasets used in this study.

Language	Source	Modality		Task					Labels			
		Text	Audio	PD	FT	SR	FC	NA	DM	AD	MCI	HC
English	Pitt	✓	✓	✓	✗	✗	✗	✗	✗	255	42	243
	Lu	✓	✓	✓	✗	✗	✗	✗	6	16	2	27
	VAS	✓	✓	✗	✗	✗	✓	✗	30	✗	35	36
	Baycrest	✓	✓	✗	✗	✓	✗	✗	✗	3	7	✗
	Kempler	✓	✓	✓	✗	✗	✗	✗	✗	7	✗	✗
	WLS	✓	✓	✓	✓	✗	✗	✗	✗	263	✗	1106
	Delware	✓	✓	✓	✗	✗	✗	✓	✗	✗	61	34
	Taukadial	✗	✓	✓	✗	✗	✗	✗	✗	✗	95	74
Spanish	Ivanova	✓	✓	✗	✗	✗	✗	✓	✗	74	90	197
	PerLA	✓	✗	✓	✓	✗	✓	✗	✗	21	✗	✗
Chinese	NCMMSE	✗	✓	✓	✓	✗	✓	✗	✗	79	93	108
	iFkyTek	✓	✗	✓	✗	✗	✗	✗	✗	68	144	111
Greek	DS3	✗	✓	✓	✓	✗	✗	✗	✗	76	✗	19
	DS5	✗	✓	✓	✓	✗	✗	✗	✗	26	35	31
	DS7	✗	✓	✓	✓	✗	✗	✗	✗	27	35	2
	ADReSS-M	✗	✓	✓	✗	✗	✗	✗	✗	22	✗	24

Dataset creation pipeline



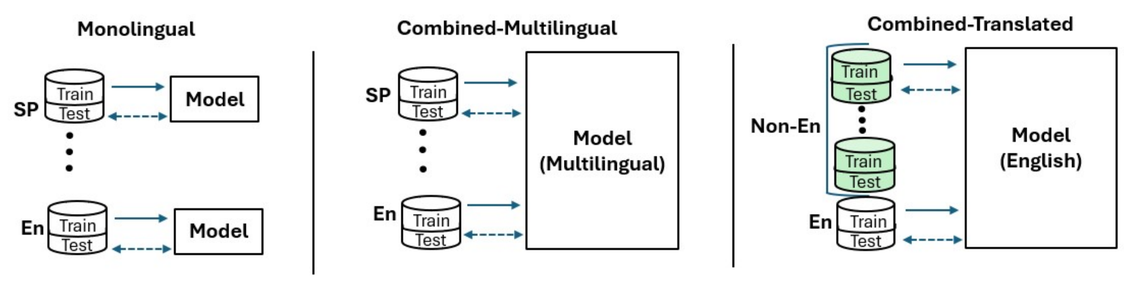
Dataset distribution

Language	Train Split			Test Split		
	AD	MCI	HC	AD	MCI	HC
Spanish	78	70	156	20	18	39
Chinese	93	200	187	23	51	47
Greek	266	51	87	66	13	22
English	512	239	1450	63	60	87
Total	**949**	**560**	**1880**	**172**	**142**	**195**

Experiment

Method pipeline

Two text representation methods, dense and sparse, and four machine learning classifiers are used separately to perform binary and three-class classification in the three following experimental scenarios.



Three-class classification results

Language	Text Repr.	Monolingual				Combined-Multilingual				Combined-Translated			
		DT	RF	SVM	LR	DT	RF	SVM	LR	DT	RF	SVM	LR
Spanish	Sparse	0.61	0.60	**0.61**	0.61	0.51 ↓	**0.62** ↑	0.51 ↓	0.58 ↓	0.47 ↓	**0.58** ↓	0.56 ↓	0.56 ↓
	Dense	0.52	0.61	**0.61**	0.61	0.47 ↓	**0.61**	0.61	0.57 ↓	**0.60** ↑	0.58 ↓	0.56 ↓	0.51↓
Chinese	Sparse	0.36	0.35	**0.40**	0.39	**0.42** ↑	0.39 ↑	0.39 ↓	0.40 ↑	0.45 ↑	0.59 ↑	**0.68** ↑	0.62 ↑
	Dense	0.51	0.58	**0.59**	0.56	0.43 ↓	**0.62**↑	0.60 ↑	0.60 ↑	0.43 ↓	**0.64**↑	0.60 ↑	0.45 ↓
Greek	Sparse	0.59	**0.74**	0.67	0.71	0.57 ↓	**0.71** ↓	0.53 ↓	0.66 ↓	0.64 ↑	0.65 ↓	**0.69**↑	0.60 ↓
	Dense	0.54	0.66	**0.73**	0.73	0.54	0.66	0.65 ↓	**0.67** ↓	**0.62** ↑	0.61 ↓	0.60 ↓	0.42 ↓
English	Sparse	0.59	0.62	**0.65**	0.65	0.59	0.58 ↓	0.41 ↓	**0.66** ↑	0.50 ↓	0.61 ↓	**0.66**↑	0.64 ↓
	Dense	0.51	0.62	**0.65**	0.63	0.50 ↓	0.62	**0.65**	0.63	0.50 ↓	0.57 ↓	**0.66** ↑	0.41 ↓

Summary

Conclusion

- Binary classification outperformed multiclass due to MCI's clinical and cognitive overlap with AD and HC.
- Chinese improved with combined-multilingual (Accuracy: 0.62) and combined-translation (Accuracy: 0.68) dataset variations; Spanish and English showed small gains, and Greek declined.
- Translation helped Chinese and English, but hurt or had no effect on Spanish and Greek, likely due to altered language-specific cues.

Challenges

- Speaker diarization remains unreliable due to noisy recordings.
- Dataset imbalance impacts classification, especially for MCI cases.
- Translation does not always preserve cognitive-linguistic signals.

Future work

- Incorporating speaker role detection for more targeted modeling.
- Developing language-specific models using interpretable linguistic analysis.
- Exploring ensemble and hierarchical deep learning architectures.
- Addressing label imbalance using advanced data augmentation methods.
- Analyzing acoustic features.

2025 – SIGIR'25, July 13–18, 2025, Padua, Italy